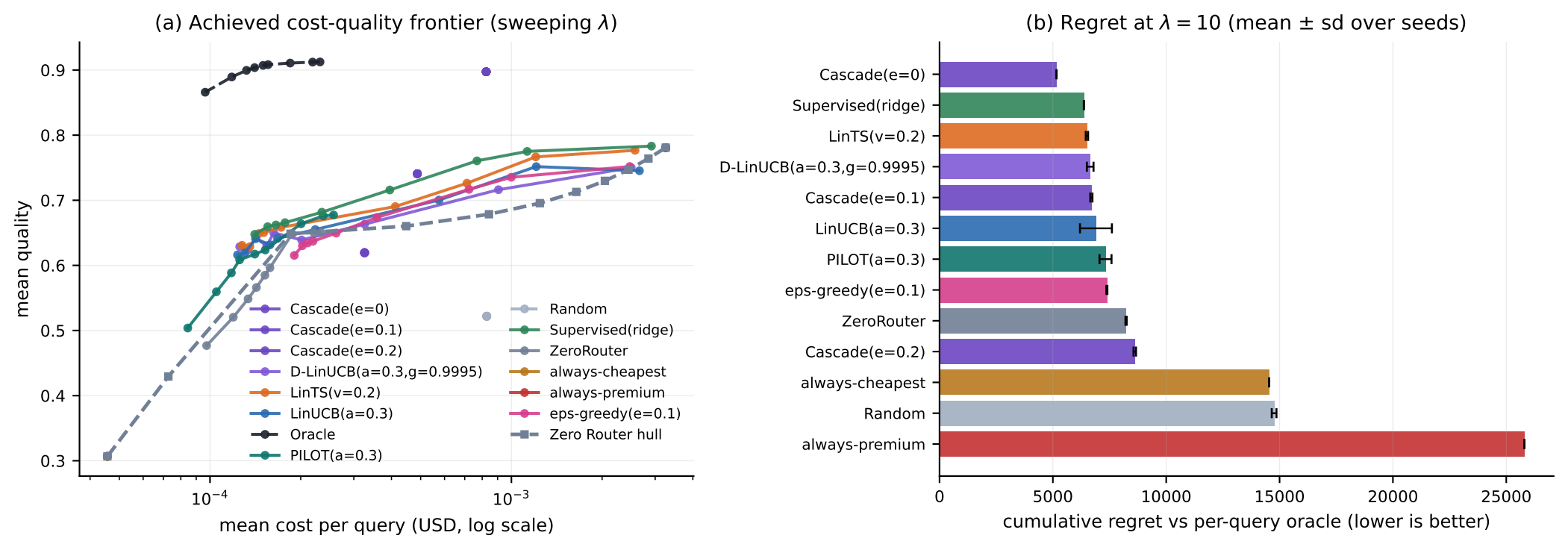
(a) Achieved cost-quality frontier (sweeping λ)
mean quality
0.9
0.8
0.7
0.6
0.5
0.4
0.3
10^{-4}
10^{-3}
mean cost per query (USD, log scale)
Cascade(e=0)
Cascade(e=0.1)
Cascade(e=0.2)
D-LinUCB(a=0.3,g=0.9995)
LinTS(v=0.2)
LinUCB(a=0.3)
Oracle
PILOT(a=0.3)
Random
Supervised(ridge)
ZeroRouter
always-cheapest
always-premium
eps-greedy(e=0.1)
Zero Router hull
(b) Regret at λ = 10 (mean ± sd over seeds)
Cascade(e=0)
Supervised(ridge)
LinTS(v=0.2)
D-LinUCB(a=0.3,g=0.9995)
Cascade(e=0.1)
LinUCB(a=0.3)
PILOT(a=0.3)
eps-greedy(e=0.1)
ZeroRouter
Cascade(e=0.2)
always-cheapest
Random
always-premium
0
5000
10000
15000
20000
25000
cumulative regret vs per-query oracle (lower is better)